CLIP Cross-Modal Image Pairing for 3D Scene Understanding

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Abstract

Reconstructing 3D scenes from unstructured image collections remains a core challenge in computer vision, particularly under real-world conditions with distractor images and ambiguous scene boundaries. Traditional pipelines such as Structure from Motion (SfM) and Multi-View Stereo (MVS) depend on robust local feature matching, but often struggle in cluttered, low-texture, or repetitive environments. In this work, we investigate the potential of CLIP—a vision-language model pretrained on large-scale image-text pairs—for improving image pair shortlisting in 3D reconstruction tasks. We propose two methods that leverage CLIP's semantic understanding: (1) a vision-only approach that uses image embeddings for clustering, and (2) a cross-modal method that constructs text-based pseudo-labels using WordNet to guide semantic grouping. These embeddings are integrated with lightweight keypoint extraction and COLMAP-based pose estimation for reconstruction. Our experiments on the Image Matching Challenge 2025 benchmark demonstrate that CLIP Vision embeddings significantly enhance clustering precision, while CLIP Cross-Modal further improves pose estimation accuracy. These findings highlight the benefits of incorporating cross-modal semantics for more robust and scalable 3D scene reconstruction. [1]

1. Introduction

Reconstructing 3D scenes from collections of unstructured images is a fundamental and longstanding challenge in computer vision. Traditional pipelines, such as Structure from Motion (SfM) and Multi-View Stereo (MVS), have achieved remarkable progress by leveraging keypoint-based feature descriptors like SIFT. However, these methods often falter in real-world conditions involving occlusions, repetitive patterns, or low-texture regions. SfM estimates camera parameters by matching sparse keypoints across views, while MVS densifies the resulting sparse point cloud by estimating per-pixel depth maps from calibrated images. The reliability of both stages hinges on accurate and robust feature matching.

The Image Matching Challenge 2025 significantly increases the difficulty of this task by introducing large, uncurated image collections containing irrelevant or visually ambiguous content. Solving this task involves not only reconstructing scenes but also identifying which images belong to the same scene—a problem analogous to assembling multiple interleaved jigsaw puzzles. Classical feature-based approaches often fail under such noisy and diverse conditions.

In this work, we investigate the use of deep visual-semantic models to address these challenges. Specifically, we explore how CLIP, a cross-modal model pretrained on large-scale image-text pairs, can generate rich semantic embeddings that aid in shortlisting image pairs for 3D reconstruction. We propose two methods that leverage CLIP's vision and text encoders to perform semantically guided image clustering and pair matching. Our pipeline integrates these embeddings with a lightweight keypoint matcher and structure-from-motion backend to assess the impact on reconstruction accuracy. Evaluations on the Image Matching Challenge 2025 dataset demonstrate the potential of CLIP-based representations in improving clustering precision, camera pose estimation, and robustness to distractors.

2. Related Work

2.1. Semantic Visual Representations and Clustering

Contrastive Language–Image Pretraining (CLIP) [8], introduced by OpenAI, has emerged as a powerful frame-

[1] Project page: https://github.com/joshhan619/CLIP_Cross_Modal_IMC2025

work for learning general-purpose visual representations by aligning image and text embeddings in a shared latent space. Unlike handcrafted features such as SIFT, which are sensitive to viewpoint or lighting changes, CLIP's vision encoder—typically a Vision Transformer (ViT) or ResNet—captures global semantic features that remain robust across diverse conditions. This semantic invariance makes CLIP embeddings particularly effective for tasks like image clustering, where semantic consistency is essential.

While most traditional clustering approaches rely solely on intrinsic visual features, recent advances have demonstrated the benefit of integrating external supervision into the clustering process. For instance, Text-Aided Clustering (TAC) introduces a novel paradigm of leveraging external textual knowledge—specifically WordNet-derived semantic labels—to guide the clustering of visual data[4]. TAC enhances feature discriminability by selecting nouns that best separate clusters and uses cross-modal mutual distillation to align image and text representations. This enables clustering that better reflects high-level semantic groupings.

In parallel, researchers have explored the use of foundation vision models such as DINOv2 to improve local feature correspondence[1]. These features embed semantic cues that can be distilled into existing keypoint descriptors, allowing for robust one-shot matching without requiring paired images at inference. By caching features and using similarity search, this approach achieves notable gains in tasks like image registration and camera localization—offering a semantic alternative to traditional learned matchers like LightGlue or LoFTR.

Together, these lines of work highlight the promise of using cross-modal semantic information—either through language supervision or foundation model features—to improve both clustering quality and visual correspondence, forming the basis for more robust 3D scene understanding.

3. Methodology

3.1. Dataset

The dataset used in this work is sourced from Kaggle's *Image Matching Challenge 2025* [2], which is specifically designed for 3D scene reconstruction from unstructured image collections. It comprises multiple datasets, each containing one or more distinct scenes. Each scene consists of a set of images annotated with ground truth camera poses, represented as a 3×3 rotation matrix (stored in row-major order) and a 3D translation vector.

To ensure a fair comparison across methods, we evaluate each one using the same training set of 1,945 images spanning 13 datasets, applying the official evaluation metrics provided by the competition.For evaluation, threshold files are provided to define acceptable distance margins, and some images are explicitly labeled as *outliers*—they do not belong to any scene and lack associated camera poses. These distractor images and multiple scenes per dataset challenge models to estimate accurate poses, group related images, and discard irrelevant ones.

3.2. Evaluation Metrics

We use the official evaluation metric from the Kaggle competition [2], which assesses performance based on:

1. **Mean Average Accuracy (mAA)**: Measures how well image clusters match ground-truth scenes.

2. **Clustering Score**: Represents the precision of image clusters relative to the correct scene assignment.

These two metrics are combined using the *harmonic mean*, analogous to the standard *F1 score*:

$$S_k = \frac{2 \times (\text{mAA} \times \text{Clustering Score})}{\text{mAA} + \text{Clustering Score}} \quad (1)$$

where:

- The **mAA score** is the proportion of correctly registered camera poses per scene.

- The **clustering score** is the fraction of images in a cluster that actually belong to the same scene.

Final scores are averaged across different datasets to obtain a single performance measure.

3.3. Experimental Setup

In this work, we evaluate the effectiveness of using a pretrained CLIP model for shortlisting image pairs in 3D scene reconstruction. We propose two methods that leverage the semantic understanding learned by CLIP.

The first method 1 uses CLIP's vision encoder, pretrained on large-scale image-text pairs, to generate semantically rich, high-dimensional embeddings for the input images. We call this method CLIP Vision. The second method 2 builds on the approach proposed in [5], where text counterparts are constructed to guide semantic clustering. This idea is inspired by CLIP's ability to perform zero-shot classification by embedding candidate class labels using its pretrained text encoder and matching them to image embeddings. In our setting, where class labels are not known a priori, we first use KMeans to identify semantic image centers in the embedding space, setting the number of clusters to $k = \lfloor N/10 \rfloor$, where N is the number of images across all datasets. Since there are 1945 images in the training set, we use $k = 48$ image semeantic centers. For each cluster center, we score a large vocabulary of nouns derived from WordNet [10] based on their similarity to the cluster embedding, and we select the top 5 ranked nouns to construct a

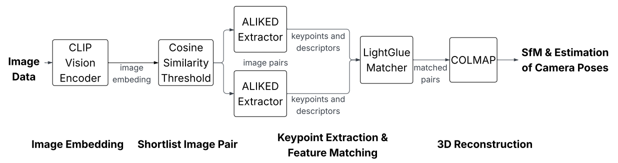

Figure 1: Overview of the proposed 3D reconstruction pipeline. CLIP's vision encoder generates semantic embeddings from input images, which are used to shortlist candidate image pairs based on cosine similarity. ALIKED extracts keypoints and descriptors for each shortlisted pair, which are then matched using LightGlue. The resulting correspondences are passed to COLMAP for Structure-from-Motion (SfM) and camera pose estimation.

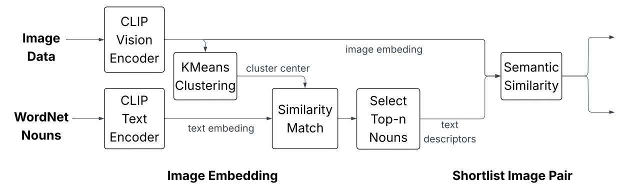

Figure 2: CLIP Cross-Modal pipeline for image pair shortlisting. Image embeddings are clustered via KMeans, and WordNet nouns are encoded using CLIP's text encoder. Text descriptors are selected based on similarity to cluster centers and used for semantic matching. The final two stages—feature extraction and 3D reconstruction—are identical to those in the CLIP vision pipeline 1.

set of semantically discriminative text descriptors. In practice, we select 230 nouns, with there being some overlap in the top 5 nouns across different semantic centers. Finally, we compute a similarity matrix between the CLIP image embeddings and these text counterparts to guide image pair shortlisting based on semantic alignment. We call this method CLIP Cross-Modal, for its use of both text and vision embeddings.

For both methods, the downstream 3D reconstruction pipeline is identical. After obtaining embeddings, we perform image pair shortlisting by matching images whose embeddings lie within a specified distance threshold. For each shortlisted image pair, we extract lightweight keypoints and descriptors using ALIKED [11], followed by feature matching with LightGlue [6]. Finally, COLMAP [9] is used for incremental structure-from-motion and pose reconstruction.

This pipeline is based on the fourth-place submission of the 2024 Image Matching Challenge [3] by user 'tmyok', where DINOv2 embeddings were used for image segmentation and shortlisting. In this paper, we use the DINOv2-based pipeline as a baseline and assess whether replacing DINOv2 with CLIP embeddings can improve clustering quality and camera pose reconstruction accuracy.

4. Analysis and Discussion

After generating text counterparts for the CLIP Cross-Modal method, we perform a sanity check by examining which counterparts were most aligned with selected example images. Figures 3 and 4 illustrate how CLIP Cross-Modal semantically labels images from the `imc2023_heritage` and `pt_stpeters_stpauls` datasets, respectively. These examples show that the text counterpart construction process enables CLIP Cross-Modal to group images from the same scene based on semantic similarity. For instance, in Figure 3, images from the 'wall' scene are consistently labeled with related terms such as 'Mycenae' and 'Mycenaean_civilization,' while outlier images receive more distinct labels. However, the counterparts are not always perfect. In Figure 4, for example, images of St. Paul's Cathedral are labeled with the name of its architect, Sir Christopher Wren, rather than the building itself. Despite these imperfections, the method still promotes meaningful separation between visually similar but semantically different locations—such as St. Paul's Cathedral and St. Peter's Square—thereby enhancing clustering quality.

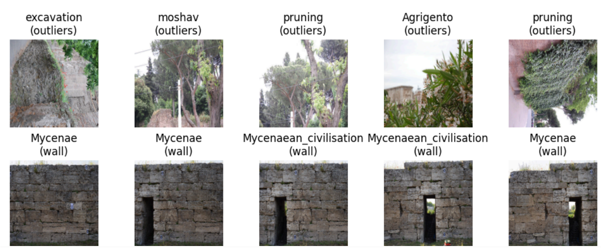

Figure 3: Top-1 text counterpart predictions for 10 images from the `imc2023_heritage` dataset. Each image is labeled with its most semantically aligned noun from the selected text counterparts, alongside its ground-truth scene label.

We further evaluate the ability of each embedding method to distinguish between scenes by visualizing their embeddings in 2D using UMAP [7] across six selected datasets, as shown in Figure 5. These visualizations highlight how the clustering difficulty varies significantly by dataset. For instance, the `stairs` dataset presents a particularly challenging case, as it contains two visually similar staircases, causing the embeddings from all three methods to map the images closely together in the embedding space. In contrast, datasets such as `pt_sacrecoeur_trevi_tajmahal` and `pt_brandenburg_british_buckingham` show clear separation of scenes across all methods. For the `imc2023_heritage` dataset, the CLIP-based em-

Figure 4: Top-1 text counterpart predictions for 10 images from the pt_stpeters_stpauls dataset. Each image is labeled with its most semantically aligned noun from the selected text counterparts, alongside its ground-truth scene label.

beddings exhibit improved clustering of the 'dioscuri' and 'wall' scenes compared to DINOv2, although they still struggle to fully distinguish 'cyprus' images from 'dioscuri'. A similar trend is observed in the stairs dataset, where the two ET scenes become more visibly separated in the CLIP-based embeddings.

Table 1 presents the performance of DINOv2, CLIP Vision, and CLIP Cross-Modal across a range of datasets, evaluated by mean Average Accuracy (mAA) and Clusterness (C). Notably, CLIP Vision achieves the highest average clusterness score of 82.76%, a massive improvement over the baseline's average clusterness score of 66.73%. This is particularly evident in datasets such as pt_brandenburg_british_buckingham, pt_stpeters_stpauls, and stairs, where CLIP Vision attains notably better clustering than DINOv2 and CLIP Cross-Modal embeddings. These gains suggest that CLIP Vision's image-only embeddings, learned from image-text pretraining, capture semantically meaningful features that are useful for image pair shortlisting.

Overall, CLIP Cross-Modal achieves the highest average mAA of 47.12%, slightly outperforming DINOv2 (46.72%) and CLIP Vision (46.22%). On specific datasets such as imc2024_lizard_pond and imc2024_dioscuri_baalshamin, CLIP Cross-Modal consistently outperforms the other methods in terms of both mAA and clusterness. However, for datasets like fbk_vineyard and stairs, all methods struggle to achieve high mAA scores, highlighting the inherent difficulty of these scenes. These results suggest that constructing text counterparts to act like pseudo labels can provide moderate improvements in shortlisting quality and camera pose estimation while still achieving a higher clustering score than the baseline.

5. Conclusion

In this study, we explored the use of CLIP-based embeddings for image pair shortlisting in 3D scene reconstruction, comparing vision-only and cross-modal approaches to a strong DINOv2 baseline. Our experiments demonstrate that CLIP Vision embeddings significantly improve the clusterness of shortlisted image pairs, achieving a 16% absolute gain over the DINOv2 baseline. This suggests that CLIP's semantic-rich representations are highly effective for grouping related images. Meanwhile, CLIP Cross-Modal embeddings yield the best overall reconstruction accuracy, suggesting that incorporating semantic text cues provides additional benefits in guiding the shortlisting process. These findings underscore the potential of leveraging pretrained vision-language models for improving downstream tasks in structure-from-motion pipelines, especially when combined with lightweight local feature matching. Future work could focus on designing more effective prompts for text counterpart construction to extract more relevant and discriminative nouns.

Dataset	**DINOv2**		**CLIP Vision**		**CLIP Cross-Modal**	
	mAA	C	mAA	C	mAA	C
imc2023_haiper	63.89	63.53	62.78	63.53	**65.56**	63.53
imc2023_heritage	27.52	**100.00**	**34.89**	98.23	26.26	98.15
imc2023_theather_imc2024_church	37.14	100.00	**40.00**	100.00	35.71	100.00
imc2024_dioscuri_baalshamin	48.84	89.41	50.46	**100.00**	**51.62**	**100.00**
imc2024_lizard_pond	45.26	**100.00**	46.26	99.06	**56.03**	**100.00**
pt_brandenburg_british_buckingham	74.65	33.33	76.50	**59.89**	**76.97**	33.33
pt_piazzasanmarco_grandplace	80.25	50.00	79.48	**100.00**	**80.71**	**100.00**
pt_sacrecoeur_trevi_tajmahal	84.72	33.33	83.56	**100.00**	**85.07**	**100.00**
pt_stpeters_stpauls	**78.99**	50.00	78.61	**99.50**	78.61	50.00
amy_gardens	14.47	100.00	12.82	100.00	**14.72**	100.00
fbk_vineyard	**20.62**	33.33	9.58	33.33	9.74	33.33
ETs	**28.8**5	43.18	23.08	**45.24**	**28.85**	43.18
stairs	2.22	71.43	2.78	77.14	2.78	50.00
Average	46.72	66.73	46.22	**82.76**	**47.12**	74.73

Table 1: Comparison of three methods across the Image Matching Challenge 2025 datasets. Metrics shown: mean Average Accuracy (mAA %) and Clusterness (C %).

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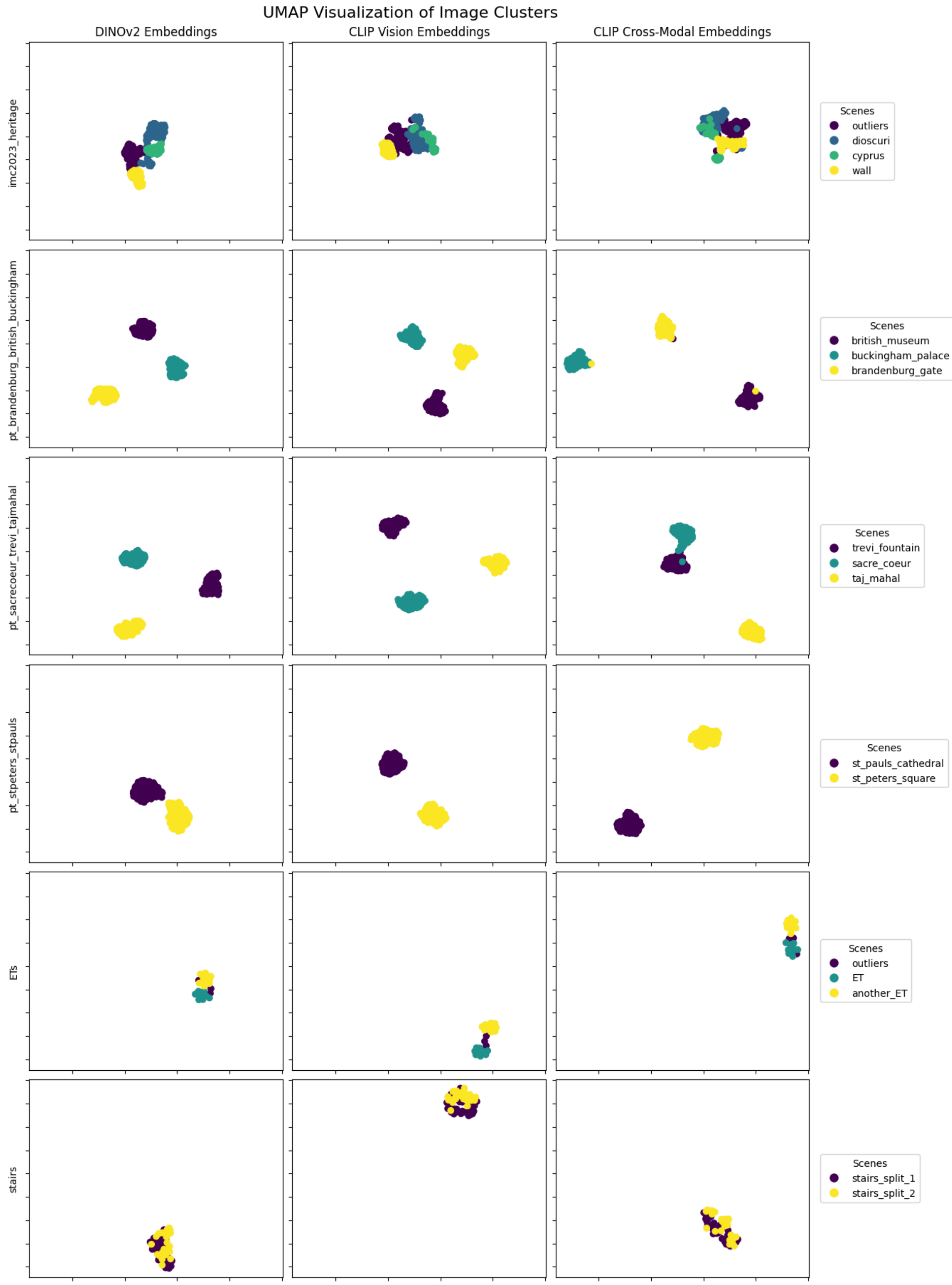

Figure 5: UMAP visualizations of the embedding spaces produced by the three evaluated methods across six selected datasets. Points are colored by ground-truth scene labels. The left column displays embeddings from the baseline DINOv2 model, the middle column shows CLIP Vision embeddings, and the right column presents CLIP Cross-Modal embeddings. We configure UMAP with 15 nearest neighbors and set the minimum distance parameter to 0.5 to balance local structure preservation with global layout.